

Bachelor in Computer Applications

Course Title: **Microprocessor and Computer Architecture**

Code No: Semester: **2**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What is microprocessor? Draw a well labelled block diagram of 8085 microprocessor and explain its different components.
2. What are different instructions available in SAP 1 computer? Explain the fetch and execution cycle of LDA instruction of SAP 1 computer along with its timing diagram.
3. What is micro-program and micro-instruction? Describe symbolic micro-program for instruction FETCH routine of basic computer.
4. Write a program to perform 8 bit division of data stored in memory location 8050 by data stored in memory location 8051 and store the quotient in memory location 8052 and remainder in 8053 memory location.
5. Explain different Logical Microoperations with example.
6. Explain the register organization of basic computer.
7. Explain memory hierarchy present in the computer system.
8. What is DMA? Explain the role of DMA controller in data transfer.
9. What is pipelining? Explain the four segment instruction pipelining.
10. Perform 2's complement multiplication of $(15) \times (-13)$ using Booth's Multiplication algorithm.
11. What is addressing mode? Explain different addressing mode in 8085 microprocessor.
12. Write short notes on: a) RISC b) Page Fault